Emergency Training Simulator POC: one-page handout

ONE-PAGE HANDOUT

Project packet / quick read for Dr. Cameron

WHAT THIS IS

- Backend-first proof of concept for a future adaptive emergency-response
- Current goal: prove the event/state/validation/log/review loop
- Original vision includes AI, Unreal, sensing, validation, immersive
- Solo-developer checkpoint toward a much larger full-system vision.
- Scope control is how the larger idea

WHAT WORKS TODAY

- FastAPI WebSocket backend receives ordered trainee actions.
- Deterministic scenario engine updates state from YAML.
- POC validation returns ok, warning, or error.
- JSONL logs support replay and post-scenario review.
- Future Unreal display path can show

CURRENT DEMO

- Run python scripts/run_local_demo.py.
- Shows happy path and warning/error path.
- Replays session logs.
- Prints review summaries that explain what happened.
- Shows environmental cue status in plain English.

CURRENT SCENARIO

- Tiny smoke-room software scenario.
- Blocked-exit hallway software scenario.
- wooden_chair_poc may block the path.
- entry_door_poc starts closed.
- Hallway obstacle and exit-door cues are tracked.

BOUNDARIES

- No real AI calls yet.
- No clinical or protocol authority.
- No generated 3D assets.
- No Unreal scene or gameplay.
- No sensors, cloud deployment, or production security yet.

NEXT DIRECTION

- Make Unreal read the generated debug text file.
- Then make Unreal parse display JSON.
- Then connect Unreal to the backend WebSocket.
- Keep the backend as the source of truth.
- Keep replay/review as the strongest proof point.

MAIN TAKEAWAY

The system can track what happened, explain it afterward, and leave room for Unreal, AI, assets, sensing, and physics later.